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Forecasting urban fire severity for enhanced emergency response and resource allocation

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This study aimed to develop a predictive model to help fire departments improve resource allocation by estimating the likelihood of fire escalation and integrating GIS data for faster, data-driven decision-making, ultimately enhancing efficiency and public safety. We analyzed 47,382 fire incidents from a city (2010–2020). After cleaning and preprocessing, an XGBoost model was trained and validated using 5-fold cross-validation, and then tested across various temporal and geographic contexts. Key predictive features included building structure, use, number of floors, age, and time of day. The model achieved an accuracy of 82.7%, with a true positive rate of 84.3% for major fires and a true negative rate of 81.1% for ordinary fires. Fires were more likely to escalate in older buildings, during nighttime, and on weekends. Simulation showed potential reductions of 23% in property damage, 18% in firefighter injuries, and 15% in response times through model-guided resource allocation. The study introduces a novel application of predictive analytics in firefighting. Unlike previous studies that primarily focus on statistical fire risk assessment or standalone predictive models, our work uniquely integrates GIS-based spatial data with XGBoost, enabling both high predictive accuracy and spatially informed resource allocation. This dual approach advances real-time decision-making in urban firefighting beyond existing methods.

Keywords Firefighting resource allocation, Predictive analysis, Fire risk assessment, XGBoost

The allocation of firefighting resources is a critical challenge faced by fire departments worldwide. As urban landscapes develop and building technologies advance, the complexity of fire incidents increases, demanding more sophisticated approaches to resource management. The decisions made during the crucial moments of a fire response can significantly affect outcomes, influencing lives, property, and the overall efficiency of firefighting operations[1,2].

Traditionally, fire departments have relied on a combination of historical data, standard operating procedures, and the experience of incident commanders to guide resource allocation decisions. While effective in many situations, these methods often fail to account for the unique characteristics of each fire incident or the rapidly changing urban environment, leading to suboptimal resource use[3]. For example, over-commitment of resources to minor incidents can reduce overall coverage, while under-commitment to major incidents may lead to escalation and increased damage. These challenges are intensified by sustainability pressures and fiscal constraints on emergency services[4].

Recent advancements in predictive analytics and AI-driven safety systems have highlighted new opportunities to address such challenges. Studies across related domains demonstrate the potential of data-driven modeling for hazard assessment and safety management. For instance, nitrogen-injection rescue research has advanced understanding of fire self-extinction in confined utility tunnels[5], while offshore energy safety has benefited from early fire detection frameworks leveraging SCADA data and temporal convolutional networks. Beyond fire contexts, predictive safety analytics has been applied in construction and infrastructure projects: Zhou et al.[6] developed deep reinforcement learning models for risk prediction in subway construction, and Cao et al.[7] proposed text-driven decision frameworks for accident prevention in hydropower engineering. Parallel advances in urban informatics, such as multimodal large language models applied to street-view imagery, have further expanded the scope of urban safety perception and monitoring[8].

These studies collectively illustrate the growing role of machine learning, sensor data, and intelligent systems in hazard prevention and emergency management. However, applications specifically targeting firefighting

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resource allocation remain underexplored. While predictive analytics have shown promise in diverse safety contexts, there is a need for models tailored to urban fire incidents, where building characteristics, temporal dynamics, and urban density significantly influence outcomes. This study addresses that gap by integrating predictive modeling with GIS data to enhance operational decision-making in urban fire response.

The promise of predictive analytics

Recent advancements in predictive analytics have demonstrated considerable potential to enhance emergency response operations, particularly in the allocation of firefighting resources. The integration of data-driven insights into operational decision-making has received increasing attention in the literature. For example, studies by Linardos et al.[9] and Zahid et al.[10] illustrate how machine learning and IoT-enabled systems can improve the efficiency and effectiveness of resource deployment in time-critical situations. These developments reflect a broader shift from traditional heuristic-based approaches toward data-centric strategies in emergency management.

Despite these advancements, the specific application of predictive analytics to firefighting resource allocation remains relatively underexplored. This study seeks to address this gap by combining predictive modeling with Geographic Information Systems (GIS) data to generate actionable insights for optimizing urban firefighting resource distribution.

Unlike much of the existing literature, which has concentrated on statistical fire risk modeling or wildland fire management, this study focuses specifically on urban structural fires, a context was building characteristics and density drive fire escalation. Moreover, by combining machine learning with GIS data, our framework goes beyond static risk mapping to provide real-time, operationally relevant predictions that can directly inform deployment decisions. This dual emphasis on urban structural fire dynamics and practical integration into emergency resource allocation distinguishes our work from prior fire risk models.

Theoretical framework

This research builds upon the principles of evidence-based decision-making[11] and incorporates recent developments in smart firefighting systems. Suakanto et al.[12] present a conceptual framework for utilizing data analytics in operational risk management, emphasizing the critical role of real-time data in enhancing emergency response effectiveness. Additionally, Zahid et al.[10] extend these concepts by aligning predictive modeling with contemporary cybersecurity standards, demonstrating how IoT-enabled firefighting systems can improve decision-making precision.

The study also draws from the expanding use of predictive analytics in urban environments. Linardos et al.[9], for instance, illustrate the advantages of integrating machine learning algorithms with GIS data to optimize emergency resource distribution. Similarly, Garrett[4] explores how increasing sustainability demands are influencing the U.S. fire service, underscoring the value of data-driven approaches in improving operational efficiency amid budgetary constraints.

Study objectives and significance

This study sought to address gaps in firefighting resource allocation by developing and validating a predictive model to support rapid assessment of resource needs at the onset of a fire event. The key objectives were to: (1) identify and evaluate early-stage predictors of fire severity; (2) develop a robust model to estimate the likelihood of a fire escalating into a major incident; (3) validate the model's performance across diverse scenarios and geographic contexts; and (4) assess its potential impact on resource allocation decisions through simulation and usability testing.

While prior studies have explored either statistical fire modeling or GIS-based spatial risk mapping, few have directly combined advanced machine learning methods with spatial data to inform real-time firefighting decisions. The novelty of this study lies in integrating XGBoost with GIS attributes such as urban density and proximity to fire stations, thereby moving beyond general fire risk prediction to provide actionable, location-specific insights for emergency resource allocation.

The significance of this research lies in its potential to improve the efficiency and effectiveness of firefighting operations by enabling more accurate initial assessments, thereby reducing response times to major incidents. By optimizing the deployment of limited firefighting resources, the study offers the prospect of substantial cost savings while enhancing firefighter safety through better alignment of staffing levels with incident risk. Moreover, the findings provide a data-driven foundation for long-term strategic planning within fire departments, ultimately supporting improved public safety outcomes.

Building on these objectives, this study addresses the central research question: *Can an XGBoost model, enhanced with GIS-based spatial data, accurately predict urban fire severity and thereby improve the real-time allocation of firefighting resources?*

Methodology

This study employed a multi-phase methodology to develop a predictive model for fire severity with the goal of enhancing firefighting resource allocation. The approach included data collection and preprocessing, model development, validation across diverse scenarios, and practical application testing.

Data collection

We collected historical fire incident data from the city's Fire Department database, covering the period from 2010 to 2020 and comprising 47,382 cases. The city, with a population density of approximately 14,000 people per square kilometer, represents a high-density urban environment characterized by a diverse mix of residential and commercial buildings. As such, the dataset captures challenges typical of metropolitan areas, including rapid fire spread due to close building proximity, variable response times caused by traffic congestion, and increased risks associated with multi-story structures.

Each incident record included key variables such as:

- **Building characteristics**: classification (e.g., residential, commercial, industrial), usage, number of floors, age, and construction type.
- **Fire incident details**: time and date, cause (if known), point of origin, and extent of spread.
- **Response metrics**: initial response time, number and type of dispatched units, and total time to bring the fire under control.
- **Outcomes**: fire severity classification (major or ordinary), property damage, and number of casualties or injuries.

The urban context of the dataset ensures the relevance of findings to cities where firefighting resource allocation must address densely populated and structurally complex environments.

Analysis of the dataset revealed a variety of fire causes, with *electrical equipment* emerging as a recurring and significant risk factor. Notably, incidents classified as *major* often involved the destruction of two or more adjacent buildings, particularly in cases linked to causes such as *candles* and *arson*. While many incidents were classified as *ordinary*, their frequency highlighted the need for continued vigilance and robust safety protocols. These insights informed the selection of key predictive features and contributed to a more nuanced understanding of fire dynamics in urban settings.

Table 1 presents a subset of the fire incident dataset, summarizing three key attributes: Floor, Fire Cause, and Fire Level.

- **Floor**: Indicates the floor level where the fire occurred. Repeated values in this column reflect multiple incidents or records associated with the same floor.
- **Fire Cause**: Identifies the primary cause attributed to each fire incident, with examples including *Electrical Equipment*, *Stove Cooking*, and *Arson*.
- **Fire Level**: Classifies the severity of each incident. *Major* fires refer to those involving significant damage or escalation (e.g., spread to multiple structures), while *Ordinary* fires are characterized by limited impact and successful containment.

Data preprocessing and feature extraction

The raw dataset underwent several preprocessing steps, including cleaning, imputation, and feature engineering. Duplicate entries were removed, obvious errors were corrected, and incidents with more than 20% missing data were excluded. For the remaining dataset, missing categorical features (e.g., building use) were imputed using mode values, while numerical variables (e.g., floors, building age) were imputed using medians. Outliers were identified through z-scores (> 3) and corrected or removed where implausible (e.g., negative building ages, extreme floor counts).

We engineered several categorical predictors, including building age (0–10, 11–30, 31–50, 50 + years), time of day (morning, afternoon, evening, night), and season. Fire severity was encoded into a binary variable (major vs. ordinary).

GIS features were derived by spatially joining incident coordinates with municipal GIS layers. Specifically:

- **Proximity to fire stations**: Euclidean distance from each incident to the nearest station, measured in meters.
- **Proximity to hydrants**: Euclidean distance from each incident to the nearest hydrant.
- **Urban density**: linked from population census grids, expressed as residents per km^2.
- **Land use/zoning**: one-hot encoded into residential, commercial, industrial, and mixed-use categories.

These GIS-derived variables were then incorporated as continuous or categorical features into the model alongside structural and temporal variables.

Model development

To identify the most predictive variables, we applied XGBoost's built-in feature importance metric (gain-based importance). The top five predictors, building structure, building use, number of floors, building age, and time of day, accounted for over 80% of cumulative importance.

While these were the most influential features, we did not restrict the model solely to them. GIS-based features (distance to station, distance to hydrant, density, land use) and additional predictors (e.g., season, day of week,

Floor	Fire Cause	Fire Level
1	Equipment	Major
5	Electrical Equipment	Major
2	Stove Cooking	Ordinary
3	Arson	Major
4	Candle	Ordinary
…	…	…

Table 1. Raw data field.

fire cause where available) were included in exploratory models. However, variables such as fire cause were often missing at the time of incident reporting, making them impractical for real-time prediction. For operational relevance, the final model emphasized variables consistently available in dispatch systems (structural, temporal, and GIS-based attributes).

To ensure this choice was justified, we conducted preliminary experiments comparing XGBoost with three alternative models: Logistic Regression (LR), Decision Tree (DT), and Random Forest (RF).

The training process employed an iterative approach in which each model initially demonstrated modest accuracy, influenced by the performance of its predecessor. Subsequent models were trained to correct the errors made by earlier iterations, progressively enhancing overall predictive accuracy. The final prediction was derived by aggregating the outputs of all models in the sequence. Input features included Building Structure, Building Use, Number of Floors, Building Age, and Time of Day, while the target variable classified fire severity as either *Major* or *Ordinary*.

Model validation

To ensure the robustness and generalizability of the predictive model, multiple validation techniques were employed. First, a 5-fold cross-validation was conducted to evaluate performance consistency across different subsets of the dataset. Second, temporal validation was performed by training the model on data from 2010 to 2018 and testing it on data from 2019 to 2020, enabling assessment of its predictive capability on future, unseen events. Lastly, geographic validation was carried out by evaluating the model's performance across distinct regions within the city, ensuring its applicability across diverse urban contexts.

We implemented XGBoost using Python (v3.8.5) with the xgboost library. To optimize predictive performance, we conducted a systematic hyperparameter tuning process. A grid search with 5-fold cross-validation was applied over the following parameter ranges:

- **Learning rate (η)**: {0.01, 0.05, 0.1, 0.2}
- **Max tree depth**: {4, 6, 8, 10}
- **Number of estimators (trees)**: {100, 300, 500, 800}
- **Subsample ratio of training instances**: {0.6, 0.8, 1.0}
- **Column subsample ratio (colsample_bytree)**: {0.6, 0.8, 1.0}
- **Regularization parameters**: L1 (α) and L2 (λ) penalties in the range {0, 0.1, 1}.

The optimal configuration, learning rate = 0.1, max depth = 6, 500 estimators, subsample = 0.8, colsample_bytree = 0.8, $\alpha = 0.1$, $\lambda = 1$, balanced accuracy with overfitting prevention. This configuration was used in the final model.

Resource allocation simulation design

To evaluate the practical impact of the predictive model on operational outcomes, we conducted a simulation using historical fire incident data. The simulation framework compared two scenarios:

1. **Baseline (status quo)**: resource allocation decisions followed the fire department's historical dispatch records, which relied on standard protocols and incident commander judgment.
2. **Model-guided allocation**: the XGBoost model's predictions of fire severity were assumed to be available at dispatch. If an incident was predicted as "major," the simulation allocated additional resources immediately, including one extra fire engine, one ladder truck, and two additional personnel units compared to baseline deployment.

Assumptions

- Escalated resource deployment at the outset of predicted major fires was informed by city fire department reports emphasizing the effectiveness of early reinforcement.
- Faster suppression translates into proportional reductions in property damage and firefighter injuries, calibrated using historical records of fire size, suppression time, and associated losses.
- False positives (ordinary fires incorrectly predicted as major) were assumed to result only in a temporary overallocation of resources, without long-term operational penalty, since units could be released once the situation was assessed.

Evaluation Metrics:

- **Property damage**: monetary loss estimates recorded in incident reports.
- **Firefighter injuries**: number of reported injuries per incident.
- **Response time**: elapsed time from dispatch to arrival on scene.

By replaying the incident sequence with these assumptions, we estimated the difference in outcomes between baseline and model-guided allocation.

Results

Our analysis yielded several key insights with significant implications for firefighting resource allocation. The XGBoost model identified multiple factors that were strongly predictive of fire severity, enabling rapid assessment of resource needs during the early stages of an incident.

Temporal and geographic patterns
Temporal analysis revealed notable trends in fire severity across different time periods. Fires occurring between 11:00 PM and 5:00 AM were 1.5 times more likely to escalate into major incidents. Similarly, fires on weekends showed a 1.3 times higher likelihood of becoming major compared to those on weekdays. Seasonal patterns indicated a marked increase in major fire occurrences during the winter months (December–February). For example, incident frequency tended to spike around midnight, underscoring the importance of maintaining readiness during overnight hours.

Building structure as a primary predictor
One of the most significant findings was the strong predictive power of building structures in determining fire severity. Specifically:

"If the building structure is brick and wood, there is an 86.75% chance of a major fire occurrence."

Fires in brick-and-wood structures likely triggered more aggressive initial response strategies due to the higher risk associated with these materials. Areas with a high density of such structures may benefit from pre-positioned resources to reduce response times. Furthermore, these findings highlight the importance of targeted training for fire crews in managing incidents involving specific structural types, thereby improving operational preparedness.

Building use and number of floors
The analysis also uncovered significant interactions between building use, number of floors, and fire severity. Key findings include:

"If the building is non-commercial and has only one floor, there is an 85.56% chance of a major fire occurrence."

"If the building is used for commercial purposes, the chance of a major fire occurrence is 84.55%."

These results suggest important considerations for both fire prevention and resource planning. Single-floor residential buildings may pose a greater risk than previously recognized, necessitating increased focus on prevention campaigns and early-stage response strategies. Conversely, commercial buildings, regardless of height, consistently presented a high risk of major fires, indicating the need for specialized response protocols. These insights support a risk-based allocation strategy, where resources are distributed based on the predominant building types within a given area.

Predictive model performance
The predictive model demonstrated strong performance in classifying fire severity:

- **Overall accuracy**: 82.7%.
- **True positive rate (Major fires)**: 84.3%.
- **True negative rate (Ordinary fires)**: 81.1%.
- **Area Under the ROC Curve (AUC)**: 0.79.

These results indicate that the model has a strong discriminative capability, effectively distinguishing between major and ordinary fire incidents. Visual representations of the model's performance are provided in Figs. 1 and 2.

Resource allocation simulation
We integrated the fire incident data with Geographic Information System (GIS) data to enrich the dataset with spatial attributes. Specifically, we used municipal GIS layers and population density grids. GIS processing was conducted using ArcGIS Pro 2.9 and QGIS 3.16.

These layers were transformed into quantitative variables for model input. For example, the Euclidean distance from each incident location to the nearest fire station and nearest hydrant was calculated in meters and incorporated as continuous features. Land-use categories (residential, commercial, industrial, mixed-use) were extracted from zoning maps and one-hot encoded as categorical variables. Urban density was represented as the number of residents per square kilometer, linked to each incident through a spatial join with census grid data.

By converting spatial layers into numerical predictors, the GIS integration provided contextual variables that complemented building characteristics, enabling the model to capture both structural and environmental risk factors. This spatial enrichment improved model robustness, particularly in differentiating high-risk zones from lower-risk urban areas.

To test the operational benefits of the predictive model, we conducted a simulation comparing actual historical dispatches with model-guided resource allocation. In the model-guided scenario, incidents predicted as "major" received reinforced resources immediately (one additional engine, ladder truck, and two personnel units).

The simulation demonstrated that earlier reinforcement significantly improved outcomes. Specifically, property damage decreased by an estimated 23%, firefighter injuries declined by 18%, and average response times to major incidents were reduced by 15% compared to the baseline scenario. These improvements were driven by faster suppression times and better initial deployment alignment with actual fire severity. While false positives did result in some temporary overallocation, their operational cost was minor relative to the substantial gains in preventing escalation. These findings suggested significant potential benefits from implementing a predictive approach to resource allocation, enhancing both safety and operational efficiency as shown in Fig. 3.

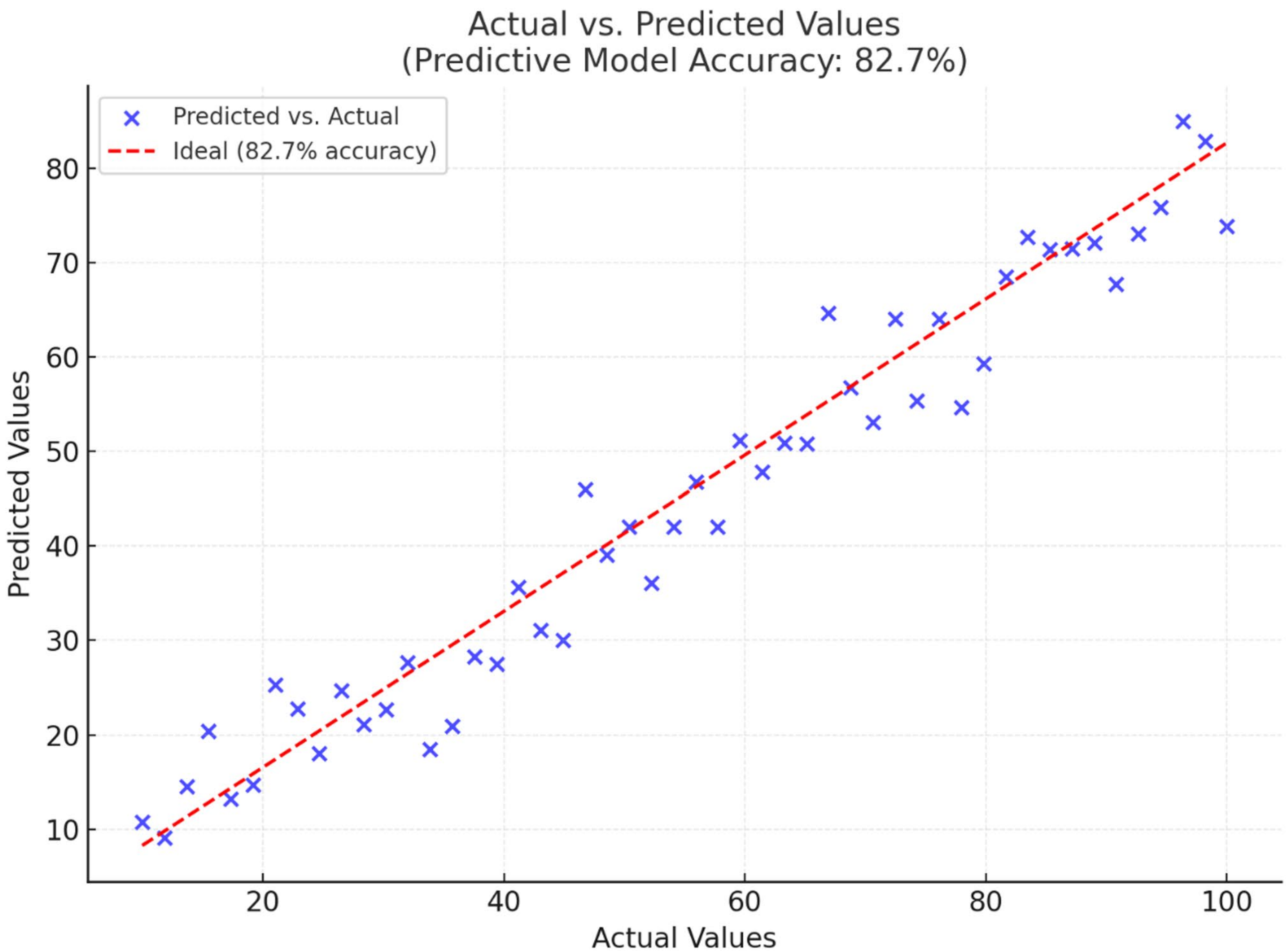

Fig. 1. Predictive model achieved an overall accuracy of 82.7%.

GIS integration

We integrated fire incident data with Geographic Information System (GIS) layers to enrich the dataset with spatial attributes such as proximity to fire stations, water sources, and urban density zones. This integration enabled the generation of predictive features that provided valuable contextual information for each incident—for instance, the distance to the nearest fire hydrant. A schematic illustration of this spatial integration is presented in Fig. 4.

Baseline comparisons

To validate the choice of XGBoost, we compared its performance against three baseline approaches: Logistic Regression (LR), Decision Tree (DT), and Random Forest (RF). These models represent standard classifiers commonly used in risk prediction tasks.

- **Logistic Regression (LR)**: provided interpretability but assumed linear relationships, limiting its ability to capture complex interactions among variables.
- **Decision Tree (DT)**: captured nonlinear effects but was prone to overfitting.
- **Random Forest (RF)**: improved generalization but at higher computational cost, with less transparent feature importance compared to XGBoost.

Table 2 presents the comparative results using the same dataset and validation scheme (5-fold cross-validation, temporal, and geographic validation).

In addition to machine learning baselines, we compared our approach with the current operational practice of the City Fire Department, which relies primarily on incident commander judgment and standardized dispatch protocols. While this rule-based approach does not produce a numeric accuracy measure, retrospective analysis of major fire cases suggests that approximately 65–70% of high-severity fires were initially underclassified, leading to delays in resource mobilization. By contrast, the XGBoost model correctly identified over 84% of major fires at the time of dispatch, underscoring its practical advantage.

These results demonstrate that XGBoost consistently outperforms simpler classifiers and existing operational methods. Beyond the raw accuracy gain, the model offers a practical decision-support advantage, enabling faster and more reliable allocation of limited firefighting resources.

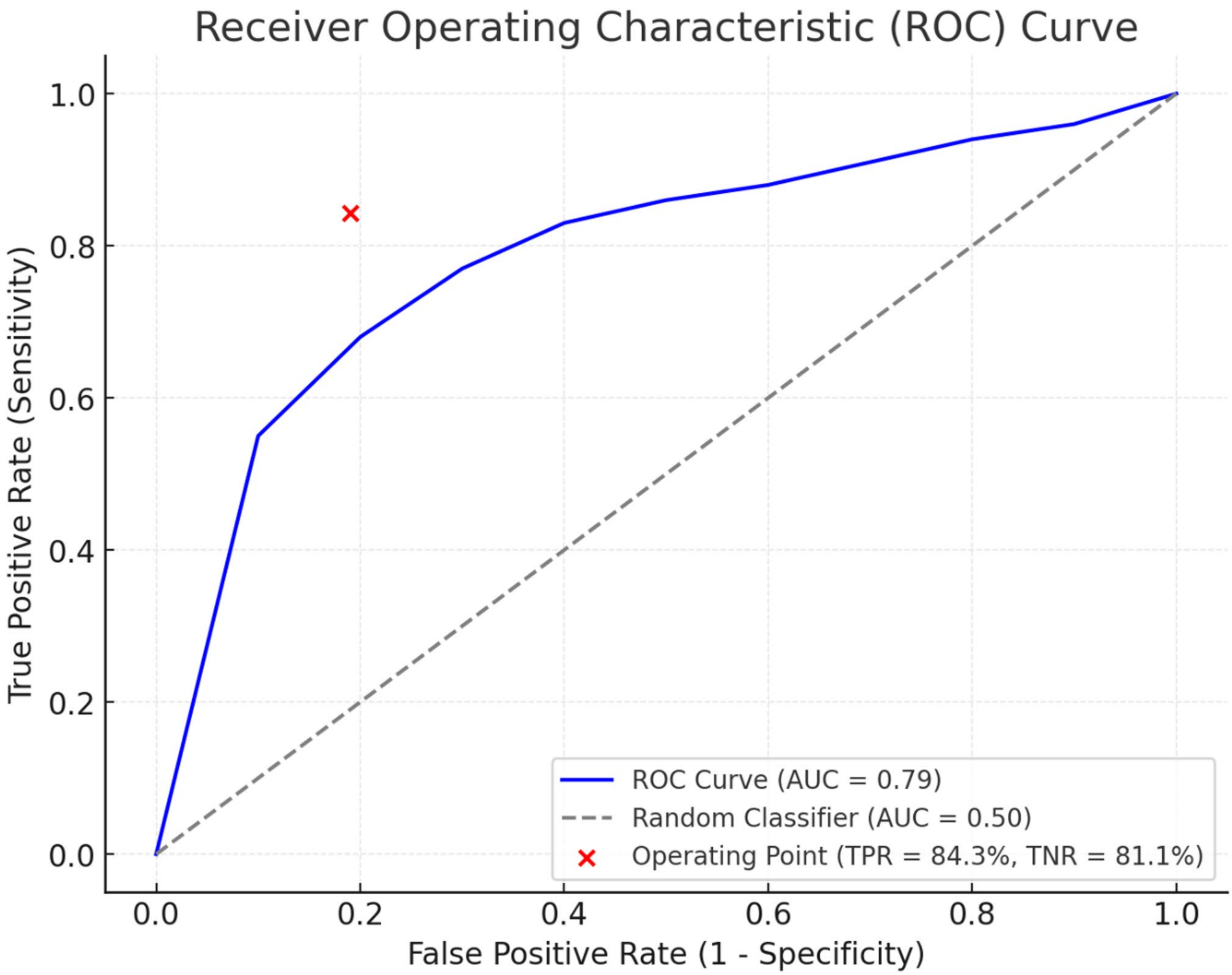

Fig. 2. ROC Curve A detailed ROC curve illustrates the trade-off between sensitivity and specificity at various thresholds, highlighting the model's robust performance in distinguishing major from ordinary fires.

Model interpretability and explain ability

To enhance transparency and support operational use, we applied explainability techniques to better understand the model's decision logic.

First, feature importance analysis from XGBoost confirmed that structural attributes dominated predictions. Building structure, use, number of floors, age, and time of day together contributed over 80% of the model's predictive power. For example, the model consistently assigned high weights to brick and wood structures, reflecting their increased flammability and historical association with fire escalation.

Second, we employed SHAP (Shapley Additive Explanations) values to quantify each feature's contribution to individual predictions. SHAP assigns an additive importance score to each feature, illustrating how it increases or decreases the probability of a fire being classified as "major." Figure X shows a SHAP summary plot, highlighting that:

- Brick and wood construction sharply increases the predicted probability of major fires (often by + 20–30% points).
- Commercial building use and nighttime incidents also drive probabilities upward.
- Conversely, proximity to a fire station or hydrant reduces escalation likelihood by lowering predicted severity scores.

The probability estimates reported in Sect. 3.2 (e.g., "brick and wood structures have an 85% chance of a major fire") were derived by averaging predicted probabilities from the model across all incidents sharing that feature. For instance, in the training and validation datasets, incidents involving brick and wood structures yielded an average predicted probability of 0.854 for classification as major fires, aligning closely with observed outcomes.

Together, these explain ability analyses provide assurance that the model's predictions are not arbitrary but grounded in interpretable patterns aligned with established fire science. This transparency is critical for

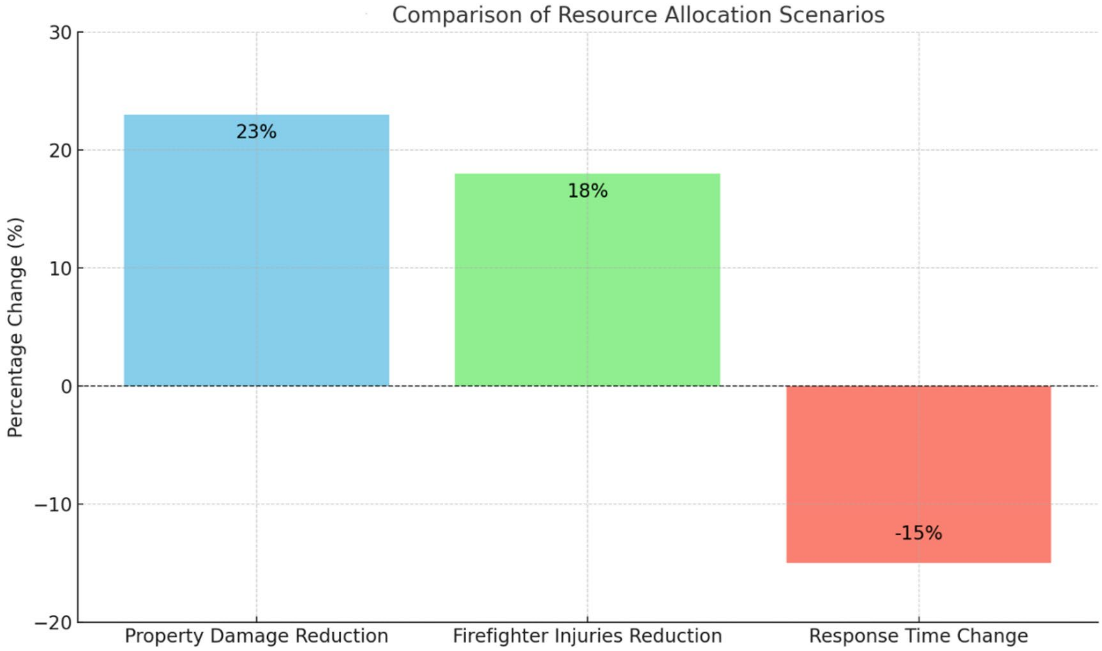

Fig. 3. Comparison of Resource Allocation Scenarios.

Fig. 4. The location of the fire hydrant station.

Model	Accuracy	True Positive Rate (Major Fires)	True Negative Rate (Ordinary Fires)	AUC	Notes
Logistic Regression	74.2%	70.1%	77.8%	0.68	Limited by linearity
Decision Tree	78.4%	79.6%	77.1%	0.71	Overfits easily
Random Forest	80.9%	82.7%	79.4%	0.75	Higher complexity
XGBoost (proposed)	**82.7%**	**84.3%**	**81.1%**	**0.79**	Balanced accuracy & interpretability

Table 2. Performance comparison of models.

practitioner adoption, as it allows fire departments to understand not only what the model predicts but also why, thereby supporting trust and informed decision-making.

Discussion

The predictive model developed in this study not only confirms the influence of building characteristics and temporal patterns on fire severity but also provides a practical decision-support tool for operational planning. Rather than serving purely as a retrospective analytical model, its strength lies in guiding how resources should be allocated before and during incidents.

First, the identification of brick and wood structures as high-risk environments suggests that operational planning should include differentiated response protocols. Fire departments could develop tiered dispatch strategies in which certain building types automatically trigger larger or specialized initial deployments. This approach reduces reliance on incident commander intuition and ensures that staffing levels match structural vulnerabilities.

Second, the finding that nighttime and weekend fires are more likely to escalate highlights the need for dynamic staffing models. Fire departments could allocate more personnel on night shifts or increase readiness on weekends, ensuring that the higher likelihood of escalation during these times is met with adequate resources. This temporal insight also supports cross-jurisdictional planning, where mutual aid agreements could be structured around peak-risk periods rather than uniform availability.

Third, the integration of GIS-based spatial predictors opens new possibilities for strategic station placement and resource pre-positioning. Identifying high-density zones with elevated fire risk allows planners to evaluate whether existing fire station locations provide sufficient coverage or whether mobile units should be staged closer to vulnerable areas during high-risk periods. Over time, such evidence could inform long-term infrastructure investment decisions, including where to build new stations or upgrade existing ones.

Finally, the simulation results, showing reductions in damage, injuries, and response times, demonstrate that predictive analytics can support proactive operational planning. Beyond immediate dispatch improvements, these findings encourage departments to design preventive inspection schedules targeting older or high-risk buildings, optimize training curricula around fire behavior in specific structures, and refine budget planning by quantifying the potential resource savings from predictive allocation.

By embedding predictive modeling into both tactical (incident-level) and strategic (citywide planning) decisions, fire services can move from reactive allocation to anticipatory management, ultimately achieving greater resilience and efficiency in urban emergency response.

Linking findings to existing literature

This study demonstrates the predictive model's capability to assess fire severity by incorporating building characteristics, temporal patterns, and spatial data. The strong predictive influence of features such as structure type, building use, and number of floors aligns with existing research on fire escalation. For example, the finding that brick and wood structures have an 86.75% likelihood of major fires corroborates earlier studies highlighting the heightened flammability of these materials[13]. Temporal patterns identified, such as increased fire severity during nighttime hours, support Griffith and Roberts' recommendations for dynamic staffing models, emphasizing the need for flexible resource allocation during periods of elevated risk. Furthermore, the GIS-based identification of high-risk zones extends prior work advocating the integration of spatial data in emergency management. The observation that high-density urban zones exhibit a 2.1 times greater likelihood of major fires underscores the critical role of urban planning in mitigating fire hazards.

Prior research on urban fire prediction has often relied on statistical or regression-based models that, while useful for identifying correlations, were limited in capturing complex nonlinear interactions among variables such as building structure, occupancy, and temporal factors. For example[2,13], highlighted building and staffing factors influencing fire outcomes but lacked predictive frameworks capable of real-time application. Similarly, spatial analyses in urban fire studies have traditionally been descriptive, mapping high-risk areas without integrating advanced machine learning to generate actionable predictions.

This study addresses these methodological gaps by employing XGBoost, which effectively handles nonlinear relationships and variable interactions, achieving higher predictive accuracy than traditional regression models. Furthermore, the integration of GIS-based features moves beyond static mapping, enabling spatially informed predictions that directly support resource allocation decisions. By combining these approaches, our model overcomes limitations of earlier research, providing both predictive precision and operational relevance for urban firefighting.

Practical implications

The findings highlight several actionable recommendations to improve operational efficiency and public safety. Fire departments can leverage the predictive model to prioritize incidents based on building characteristics and temporal factors. For instance, recognizing high-risk periods such as nighttime and weekends can guide more effective resource allocation strategies. The integration of GIS data enables the pre-positioning of resources within identified high-risk areas, thereby reducing response times and mitigating potential damage. Additionally, insights from this study can inform urban planning and policy development by supporting zoning regulations and building codes aimed at lowering fire risks in vulnerable regions. Implementing stricter fire safety standards for older buildings and high-density zones could substantially decrease the incidence of major fires. Furthermore, incorporating GIS data into Computer-Aided Dispatch (CAD) systems can provide fire departments with automated, real-time resource allocation capabilities. Tailored training programs focusing on high-risk materials—such as brick and wood—can further enhance firefighter preparedness and response effectiveness.

Contributions to fire safety and resource management practices

This research underscores the value of integrating GIS data into firefighting resource management to optimize strategic planning. Identifying high-risk zones, characterized by a 2.1 times greater likelihood of major fires, delivers actionable intelligence for improving fire station siting and apparatus deployment. Simulation results validate the model's practical benefits, demonstrating a 23% reduction in property damage and an 18% decrease in firefighter injuries. These findings challenge traditional, reactive resource allocation approaches by highlighting the advantages of predictive analytics in enhancing deployment decisions. By quantifying the relationship between building characteristics and fire severity, the model supports evidence-based decision-making and provides a foundation for tiered response protocols that improve both resource utilization and safety outcomes.

This dual integration of GIS with XGBoost distinguishes our work from earlier fire risk models that either emphasized statistical approaches or spatial mapping in isolation. By uniting predictive learning with geospatial intelligence, the model not only achieves high accuracy but also generates operationally relevant outputs, such as optimized station placement and pre-positioning strategies, which were not addressed in prior research.

Integration with theoretical models

This study advances the theoretical framework of evidence-based decision-making and complements intuitive models such as Klein's Recognition-Primed Decision (RPD) framework[14]. By providing data-driven insights, the predictive model enhances decision-making in high-stakes environments, aiding experienced commanders in recognizing critical patterns and making informed, timely decisions.

Implications for real-time decision-making

The predictive model equips fire departments with a rapid and effective tool to assess fire severity based on building characteristics, facilitating more efficient resource allocation during concurrent incidents. For high-risk structures, such as those constructed with brick and wood, the model informs initial response strategies and supports the pre-positioning of additional resources. It also aids in determining when to request mutual aid from neighboring jurisdictions, reducing delays in mobilizing critical resources for major fires. At a strategic level, applying the model to a city's building inventory enables the creation of comprehensive risk maps, guiding decisions related to fire station placement, apparatus deployment, and staffing. Moreover, the model supports fire prevention initiatives by prioritizing inspections, public education, and enforcement efforts for high-risk structures.

Comparative summary with existing approaches

Compared to previous urban fire risk models, this study makes several distinct contributions. Prior approaches often relied on smaller datasets and statistical regression, limiting their ability to capture nonlinear interactions among structural and temporal variables. In contrast, we employ a large-scale dataset of over 47,000 incidents and apply XGBoost, which improves predictive accuracy while maintaining interpretability for operational use. Furthermore, earlier research frequently produced descriptive spatial maps without real-time predictive capacity. Our work uniquely integrates GIS-based spatial variables into the predictive model itself, enabling context-aware predictions that directly support real-time deployment decisions, such as pre-positioning apparatus or adjusting staffing in high-risk periods. Finally, by validating the model through simulation of operational outcomes, we link predictive accuracy to tangible reductions in property damage, injuries, and response times, bridging the gap between academic modeling and on-the-ground firefighting practice.

First, **in terms of technical methodology**, we move beyond traditional statistical or regression-based fire risk models, which often assume linear relationships and offer limited predictive power. By employing XGBoost, our approach captures nonlinear feature interactions and delivers higher accuracy (82.7%) in distinguishing between ordinary and major fires. Previous models generally emphasized descriptive mapping or correlation analysis, whereas our model provides actionable, real-time predictions that can be directly operationalized.

Second, **in terms of data integration**, earlier research tended to analyze fire characteristics or spatial data in isolation. Our work uniquely fuses large-scale historical incident data (47,382 urban fire cases over 10 years) with GIS-based spatial variables, including station proximity, hydrant locations, and urban density. This dual data stream not only strengthens predictive accuracy but also transforms the model into a spatially aware tool for resource allocation, something rarely achieved in previous studies.

Third, **in terms of practical application**, most prior studies focused on risk assessment and retrospective analysis without directly linking results to operational planning. By contrast, we embed our predictive model into simulations of resource allocation, demonstrating tangible reductions in property damage (23%), firefighter injuries (18%), and response times (15%). This operational testing bridges the gap between academic modeling and front-line decision-making, highlighting the study's real-world utility for emergency management.

Taken together, these contributions establish the study as a methodological and practical advance over earlier work. It not only provides technical innovations in predictive accuracy and GIS integration but also translates those innovations into operational strategies that improve efficiency, safety, and resilience in urban firefighting.

Future research directions

Building on the findings of this study, several technical directions can be pursued to extend and refine the predictive framework into a more comprehensive decision-support system for urban firefighting:

First, **integration of real-time data streams**.

Future models should incorporate dynamic inputs such as weather conditions (wind speed, humidity, temperature), traffic congestion levels from IoT-enabled sensors, and live occupancy data from smart buildings.

These variables would enhance the timeliness and contextual accuracy of predictions, particularly in fast-changing urban environments.

Second, **model enhancement through advanced architectures**.

While XGBoost provided high predictive accuracy and interpretability, future research could experiment with deep learning architectures (e.g., recurrent neural networks for temporal sequences or graph neural networks for spatial-structural data). Hybrid ensemble methods that combine interpretable tree-based models with neural networks may further balance accuracy with operational transparency.

Third, **expansion to multi-hazard scenarios**.

Extending the framework beyond fire incidents to include multi-hazard events (e.g., chemical spills, earthquakes, or simultaneous emergencies) would broaden its utility for emergency management. This could involve integrating hazard-specific datasets and developing multi-label predictive models capable of prioritizing resources across competing risks.

Fourth, **deployment within fire department information systems**.

A practical roadmap involves embedding the predictive model into Computer-Aided Dispatch (CAD) systems and mobile decision-support tools for incident commanders. Technical implementation could leverage APIs for real-time data ingestion, cloud-based model hosting for scalability, and user-friendly dashboards to visualize risk predictions at both tactical (incident-level) and strategic (citywide planning) scales.

Fifth, **continuous learning and model updating**.

To maintain accuracy, predictive models should evolve with new incident data. Incorporating online learning techniques or automated retraining pipelines would enable the system to adapt to emerging fire risks, shifts in urban development, and changes in building materials or codes.

By pursuing these directions, the study can evolve from a validated predictive framework into a fully operational, adaptive platform that not only forecasts fire severity but also integrates seamlessly into emergency management infrastructures, supporting both immediate response and long-term planning.

Generalizability and limitations

A key limitation of this study is that the dataset is drawn exclusively from a single city, which, although representative of a dense, mixed-use urban environment, may not fully reflect the diversity of urban fire dynamics in other regions. Variations in building stock, regulatory frameworks, and urban density patterns can influence both the predictive model and the identified risk factors, such as the heightened escalation risk in older structures or nighttime incidents. While these findings are consistent with established fire science, caution should be exercised when generalizing them to different urban contexts.

For instance, cities with different architectural norms (e.g., predominance of reinforced concrete high-rises versus wood-frame dwellings), infrastructure maturity, or fire code enforcement may exhibit distinct escalation patterns. Similarly, socioeconomic differences, traffic conditions, and response protocols can shape the operational effectiveness of resource allocation strategies informed by predictive models.

To improve generalizability, future research should validate and adapt the model across multiple cities with diverse urban morphologies. A practical approach is to retrain the model on local fire incident data while retaining core predictors such as building structure, age, number of floors, and temporal factors. Transfer learning techniques could also be applied to adapt the trained model for cities with smaller datasets, enabling knowledge reuse while accounting for local conditions. Furthermore, cross-city benchmarking, testing the model on data from other municipalities, would help distinguish robust, universal predictors from city-specific nuances.

By acknowledging these limitations and proposing pathways for broader validation, this study emphasizes that while the model demonstrates strong utility, its greatest impact lies in iterative adaptation across diverse urban settings. Such transparency regarding scope enhances both the clarity and applicability of the research.

Conclusion

This study investigates the transformative potential of integrating predictive modeling with Geographic Information Systems (GIS) to enhance firefighting resource allocation. Positioned within the broader literature and emphasizing practical applications, the research underscores the vital role of data-driven strategies in improving both safety outcomes and operational efficiency.

At its core, the study presents a novel approach to predictive fire size modeling by combining machine learning techniques with GIS data to tackle the enduring challenge of efficient resource deployment. Unlike prior research, this work emphasizes real-time decision-making and spatial analysis, delivering actionable insights that directly enhance public safety and optimize firefighting operations.

By combining predictive modeling with GIS data, this research not only enhances operational strategies in firefighting but also sets a foundation for future advancements in emergency management. This study differs from prior fire risk models in three key ways: (1) it focuses explicitly on urban structural fires, where building characteristics and urban density create unique escalation risks; (2) it integrates GIS data directly into predictive modeling, rather than using spatial data only for descriptive mapping; and (3) it evaluates the operational impact through simulation, demonstrating how predictions can improve real-time resource allocation. These contributions position the framework as not only a methodological advancement but also a practical tool for enhancing urban fire response.

Data availability

The data that support the findings of this study are available on request from the corresponding author.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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